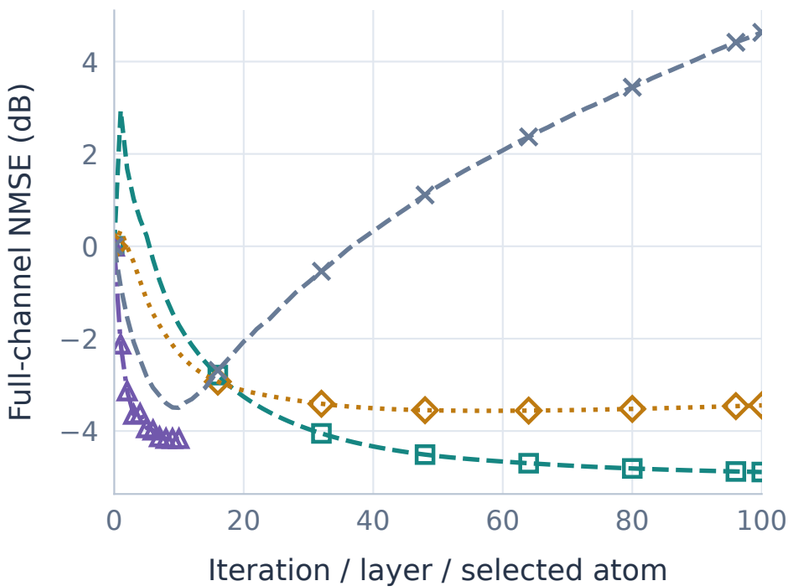
Convergence on off-grid AFDM channels

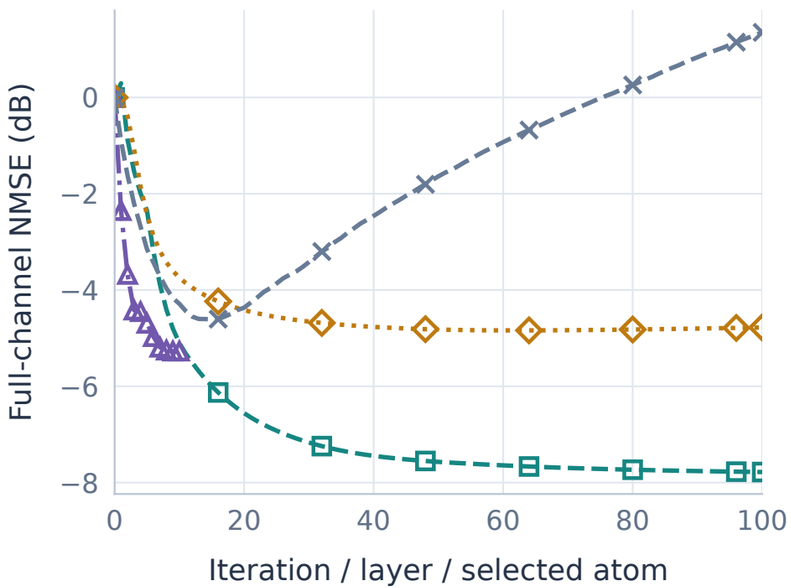
40 candidate users | 16 pilot frames | 16 receive antennas | Activity probability = 0.1

Off-grid SBL (n=20)
GNN-ASPL (n=20)
GAMP-SBL (n=20)
SOMP (n=20)

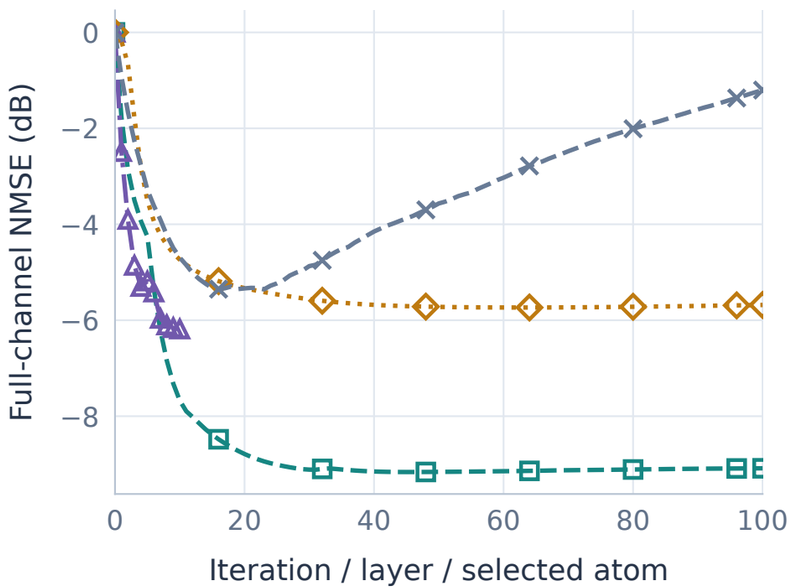
SNR = -4 dB



SNR = 0 dB



SNR = 4 dB



n = paired physical scenes per SNR. Estimated phase and noise.

Curves follow the saved trajectories and actual iteration counts. Stopped scenes retain their final estimates.

Iteration counts are not equal computation budgets; lower NMSE is better.